

NLP Coding Total 2 Assignment in this single Colab file

Load the Data

```
import pandas as pd

train_df = pd.read_excel("/content/english-hindi-train.xlsx")
valid_df = pd.read_excel("/content/english-hindi-valid.xlsx")
```

Word count Dataset

```
df = pd.concat([train_df, valid_df])
print(df.head)
print(df.shape)
print(df.info())
print("Total Row", df.shape[0])
```

```
<bound method NDFrame.head of
0      3- The matter has been referred to the Full Be...
1      4- In Reeta Mishra & Ors. vs. Director , E...
2      it is not in order to assert a legal right , b...
3      a writ of mandamus can truly be claimed in the...
4      it is not to be granted where , in fact , befo...
...
4995  Mr. Singh has also relied upon the opinion mad...
4996  in the context of the finding arrived at by th...
4997  it was lastly submitted that the circumstances...
4998  15 . I have heard learned counsels for the par...
4999  16- The compensation payable under Section 124...
```

	Source	Translation	ID
0	3 - मामला इससे संबंधित प्रश्न पर पूर्ण पीठ को ...	NaN	
1	4 - रीता मिश्रा एवं अन्य बनाम निदेशक , प्राथमि...	NaN	
2	यह किसी वैधानिक अधिकार को सिद्ध करने के लिए नह...	NaN	NaN
3	किसी वैधानिक सार्वजनिक दायित्व के स्पष्ट , सीम...	NaN	
4	इसे तब प्रदान नहीं किया जाना है जहां इसे परिवर...	NaN	
...	...	...	...
4995	श्री सिंह ने पैरा 10 में उपस्थित अधिकरण द्वारा...	54996.0	
4996	निर्णय तथा पंचाट के पैरा 13 में अधिकरण द्वारा ...	54997.0	
4997	अन्त में यह निवेदन किया गया कि अधिकरण द्वारा ध...	54998.0	
4998	15 . मैंने पक्षकारों के विद्वान अधिवक्ताओं को ...	54999.0	
4999	16 . अधिनियम की धारा 124A के अधीन देय प्रतिकर ...	55000.0	

```
[55000 rows x 3 columns]>
(55000, 3)
<class 'pandas.core.frame.DataFrame'>
Index: 55000 entries, 0 to 4999
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Source      55000 non-null  object
1   Translation  55000 non-null  object
2   ID          5000 non-null   float64
dtypes: float64(1), object(2)
memory usage: 1.7+ MB
None
Total Row 55000
```

Word Count Analysing

```
df['english_word_count'] = df['Source'].apply(lambda x: len(str(x).split()))
df['hindi_word_count'] = df['Translation'].apply(lambda x: len(str(x).split()))
```

```
df[
    (df['english_word_count'] >= 50) &
    (df['english_word_count'] <= 80)
]
```

	Source	Translation	ID	english_word_count	hindi_word_count
33	learned counsel has also relied upon the verdi...	विद्वान अधिवक्ता ने इसी प्रतिपादना के लिये अनी...	NaN	62	69
34	5- Learned counsel for the State submitted tha...	5 - राज्य के विद्वान अधिवक्ता ने निवेदन किया क...	NaN	59	63
37	more than that , in the case of Anita Prakash ...	इससे भी बढ़कर , अनीता प्रकाश (ऊपर) के मामले ...	NaN	54	50
38	moreover , the case of Smt . Rani Kumari (sup...	इसके अतिरिक्त श्रीमती रानी कुमारी (ऊपर) का म...	NaN	60	63
40	thus , in the considered opinion of the Court ...	इस प्रकार , न्यायालय की सुविचारित राय में , पू...	NaN	50	45
...	...	...	...	...	...
4988	learned counsel also makes a reference to the ...	विद्वान अधिवक्ता पुलिस द्वारा प्रस्तुत अन्तिम ...	54989.0	57	55
4989	it is submitted that in the accident , the inf...	यह निवेदन किया गया है कि दुर्घटना , दुर्घटना म...	54990.0	64	69
	R- It is submitted that these relevant	R - यह निवेदन किया गया है कि यह सम्बन्धित			

Word count Calculation

```
df["Word_Diff"] = df["english_word_count"] - df["hindi_word_count"]
```

```
df = df[df["Word_Diff"].between(-10, 10)]
```

```
df["Character_Count_English"] = df["Source"].astype(str).apply(len)
df["Character_Count_Hindi"] = df["Translation"].astype(str).apply(len)
```

```
df = df.head(10000)
```

```
df.to_excel("Final_Cleaned_English_Hindi_Dataset.xlsx", index=False)
```

```
print(df.head())
```

```

Source \
0 3- The matter has been referred to the Full Be...
1 4- In Reeta Mishra & Ors. vs. Director , E...
2 it is not in order to assert a legal right , b...
3 a writ of mandamus can truly be claimed in the...
4 it is not to be granted where , in fact , befo...

Translation ID english_word_count \
0 3 - मामला इससे संबंधित प्रश्न पर पूर्ण पीठ को ... NaN 41
1 4 - रीता मिश्रा एवं अन्य बनाम निदेशक , प्राथमि... NaN 39
2 यह किसी वैधानिक अधिकार को सिद्ध करने के लिए नह... NaN 18
3 किसी वैधानिक सार्वजनिक दायित्व के स्पष्ट , सीम... NaN 19
4 इसे तब प्रदान नहीं किया जाना है जहाँ इसे परिवर... NaN 30

hindi_word_count Word_Difference Word_Diff Character_Count_English \
0 47 -6 -6 229
1 47 -8 -8 185
2 22 -4 -4 73
3 29 -10 -10 109
4 31 -1 -1 138

Character_Count_Hindi
0 237
1 210
2 100
3 146
4 149

```

Assignment 2 Translation On LLM

```
!pip install transformers accelerate sentencepiece sacrebleu pandas openpyxl
```

[Show hidden output](#)

```
df = pd.read_excel("/content/Final_Cleaned_English_Hindi_Dataset.xlsx")
```

```
print(df.head)
```

```
<bound method NDFrame.head of
0      3- The matter has been referred to the Full Be...
1      4- In Reeta Mishra & Ors. vs. Director , E...
2      it is not in order to assert a legal right , b...
3      a writ of mandamus can truly be claimed in the...
4      it is not to be granted where , in fact , befo...
...
9995  Hon &apos;ble Aditya Kumar Trivedi , J. Nirmal...
9996  he also stated that since he had not provided ...
9997  he resisted her , in which Nirmal Das infligte...
9998  during this time , co-accused Ajay Das and Con...
9999  her husband , who had gone to a shop situated ...

                                Translation ID \
0      3 - मामला इससे संबंधित प्रश्न पर पूर्ण पीठ को ... NaN
1      4 - रीता मिश्रा एवं अन्य बनाम निदेशक , प्राथमि... NaN
2      यह किसी वैधानिक अधिकार को सिद्ध करने के लिए नह... NaN
3      किसी वैधानिक सार्वजनिक दायित्व के स्पष्ट , सीम... NaN
4      इसे तब प्रदान नहीं किया जाना है जहां इसे परिवर... NaN
...
9995  माननीय आदित्य कुमार त्रिवेदी , न्यायमूर्ति निर... NaN
9996  उसने यह भी कथित किया था कि चूंकि उसने रास्ता उ... NaN
9997  उसने प्रतिरोध किया था जिसपर निर्मल दास ने उसकी... NaN
9998  इसके दौरान , सह - अभियुक्त अजय दास तथा कांग्रेस... NaN
9999  उसका पति , जो निकट अवस्थित किसी दुकान पर गया थ... NaN

    english_word_count  hindi_word_count  Word_Difference  Word_Diff \
0                    41                 47                -6         -6
1                    39                 47                -8         -8
2                    18                 22                -4         -4
3                    19                 29               -10        -10
4                    30                 31                -1         -1
...
9995                  30                 29                 1          1
9996                  22                 23                -1         -1
9997                  29                 29                 0          0
9998                  18                 24                -6         -6
9999                  24                 26                -2         -2

    Character_Count_English  Character_Count_Hindi
0                        229                      237
1                        185                      210
2                         73                      100
3                       109                      146
4                       138                      149
...
9995                     142                      147
9996                     106                      103
9997                     133                      128
9998                     102                      109
9999                     119                      111

[10000 rows x 9 columns]>
```

```
sample_df = df.sample(200, random_state=42)
```

```
english_sentences = sample_df["Source"].tolist()
reference_hindi = sample_df["Translation"].tolist()
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
from huggingface_hub import login

try:
    login()
except Exception as e:
    print(f"Could not log in to Hugging Face Hub: {e}")

model_name = "Helsinki-NLP/opus-mt-en-hi"

# Load tokenizer and model directly
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

# Create a translation function
def translate_text(text):
    inputs = tokenizer(text, return_tensors="pt", padding=True, truncation=True)
```

```
outputs = model.generate(**inputs)
translated_text = tokenizer.decode(outputs[0], skip_special_tokens=True)
return translated_text

print(f"Model '{model_name}' loaded successfully and translation function created.")
```

Loading weights: 100% 258/258 [00:00<00:00, 6479.55it/s]
Model 'Helsinki-NLP/opus-mt-en-hi' loaded successfully and translation function created.

```
generated_hindi = []

for sentence in english_sentences:
    translated = translate_text(sentence)
    generated_hindi.append(translated)

print(generated_hindi[:5])
```

[Show hidden output](#)

```
from sacrebleu import corpus_bleu, corpus_chrf, corpus_ter

bleu = corpus_bleu(generated_hindi, [reference_hindi])
chrf = corpus_chrf(generated_hindi, [reference_hindi])
ter = corpus_ter(generated_hindi, [reference_hindi])

print("BLEU:", bleu.score)
print("CHRF:", chrf.score)
print("TER:", ter.score)
```

Upto Translate into Hindi complited An In Github main i wrote my Work on this project using tools